

An elastic network model with $k_{ij} = k(l_{ij})$: potential, conformational distribution, and the effect of mutation

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An elastic network model is specified by the force constants k_{ij} and the equilibrium lengths l_{ij} of its springs. Here the force constants are taken to be a function of the equilibrium lengths, $k_{ij} = k(l_{ij})$, which leaves the l_{ij} as the only free parameters. A mutation perturbs the l_{ij} of one site, and its consequences are obtained without expanding or linearising the model. This note derives the potential, its minimum, the conformational distribution, and the thermodynamic properties, first for a general network and then for a reference and its mutant, ending with the mutational changes.

Contents

1	The model	2
1.1	Conformations and distances	2
1.2	Parameters	2
1.3	The potential	2
2	The minimum of the potential	3
2.1	The equilibrium condition	3
2.2	Strain	3
2.3	Frustration	4
3	The conformational distribution	4
3.1	The Boltzmann distribution	4
3.2	The Hessian	4
3.3	Longitudinal and transverse stiffness	5
3.4	The Gaussian approximation	5
4	Thermodynamic properties	6
4.1	The partition function	6
4.2	Free energy, internal energy, entropy	6
5	Reference and mutant	7
5.1	The reference network	7
5.2	Mutation	7

6	Mutational changes	8
6.1	Three energy differences	8
6.2	Decomposition of the stress energy	8
6.3	The three terms	9
6.4	Differences and energies	9
6.5	Changes in the thermodynamic properties	10
7	The step function	10
8	Open questions and work to come	11

1 The model

1.1 Conformations and distances

The protein is represented by N nodes. A conformation is the set of their positions,

$$\mathbf{r} = (\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) \in \mathbb{R}^{3N}, \quad (1)$$

and the distance between nodes i and j in conformation $\mathbf{r}$ is

$$d_{ij}(\mathbf{r}) = \|\mathbf{r}_i - \mathbf{r}_j\| = [(\mathbf{r}_i - \mathbf{r}_j) \cdot (\mathbf{r}_i - \mathbf{r}_j)]^{1/2}. \quad (2)$$

We write $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ throughout, so that $\mathbf{r}_{ij} = -\mathbf{r}_{ji}$ and $d_{ij} = d_{ji}$.

1.2 Parameters

Each pair of nodes carries an equilibrium length l_{ij} , the length at which the spring joining the pair is unstretched. There is one for every pair,

$$\Lambda = \{l_{ij} : 1 \leq i < j \leq N\}, \quad |\Lambda| = \frac{1}{2}N(N-1), \quad (3)$$

and Λ is the complete set of free parameters of the model. The force constant of a pair is fixed by its equilibrium length through a function k ,

$$k_{ij} = k(l_{ij}), \quad k : [0, \infty) \rightarrow [0, \infty). \quad (4)$$

This is the one departure from standard practice. A standard elastic network model is built on a given structure $\mathbf{r}^e$ and takes $k_{ij} = f(d_{ij}(\mathbf{r}^e))$, a function of a distance in that structure; here k is a function of a free parameter of the model. The function k is otherwise arbitrary. We take it non-increasing, so that pairs with long equilibrium lengths are weakly coupled or uncoupled, and we assume $k(l) = 0$ for l large enough that distant pairs do not interact.

1.3 The potential

The potential energy of a conformation $\mathbf{r}$ for a network with parameters Λ is a sum of harmonic terms, one per pair,

$$V(\mathbf{r}; \Lambda) = \frac{1}{2} \sum_{i < j} k(l_{ij}) (d_{ij}(\mathbf{r}) - l_{ij})^2. \quad (5)$$

The sum runs over all $\frac{1}{2}N(N-1)$ pairs. Since $V \geq 0$ and $V = 0$ only if every term vanishes, the zero of energy is the fully relaxed configuration, and V evaluated at any conformation is the elastic energy stored in the springs.

A pair with $k(l_{ij}) > 0$ contributes to V and is said to be *active*; a pair with $k(l_{ij}) = 0$ contributes nothing. The set of active pairs is the network's contact map. Because k is evaluated at l_{ij} and not at $d_{ij}(\mathbf{r})$, which pairs are active is fixed by Λ alone, and does not depend on the conformation at which V is evaluated.

Every pair carries a parameter, including the inactive ones. This is necessary if a mutation is to be able to create a contact: an inactive pair with no l_{ij} could never acquire one inside the range where $k > 0$, so mutation could only destroy contacts and never form them.

2 The minimum of the potential

2.1 The equilibrium condition

To find the conformations that minimise (5) we need the gradient of V , and hence the gradient of the distances. Differentiating $d_{ij}^2 = \mathbf{r}_{ij} \cdot \mathbf{r}_{ij}$ with respect to $\mathbf{r}_i$,

$$2 d_{ij} \frac{\partial d_{ij}}{\partial \mathbf{r}_i} = 2 \mathbf{r}_{ij}, \quad \text{so} \quad \frac{\partial d_{ij}}{\partial \mathbf{r}_i} = \frac{\mathbf{r}_{ij}}{d_{ij}} \equiv \hat{\mathbf{e}}_{ij}, \quad (6)$$

which defines the unit vector $\hat{\mathbf{e}}_{ij}$ along the pair, pointing from j to i . Since d_{ij} depends on $\mathbf{r}_j$ only through $\mathbf{r}_{ij} = -\mathbf{r}_{ji}$, we also have $\partial d_{ij} / \partial \mathbf{r}_j = -\hat{\mathbf{e}}_{ij}$.

Differentiating (5) by the chain rule, and noting that $k(l_{ij})$ is a constant with respect to $\mathbf{r}$,

$$\frac{\partial V}{\partial \mathbf{r}_i} = \sum_{j \neq i} k(l_{ij}) (d_{ij}(\mathbf{r}) - l_{ij}) \hat{\mathbf{e}}_{ij}. \quad (7)$$

Each term is a central force along the line joining the pair. That k does not differentiate is a consequence of (4): had we taken $k_{ij} = f(d_{ij}(\mathbf{r}))$, a term in $f'(d_{ij})$ would appear here and again in every derivative below.

An equilibrium conformation $\mathbf{r}^e$ is one at which the gradient vanishes,

$$\boxed{\sum_{j \neq i} k(l_{ij}) (d_{ij}(\mathbf{r}^e) - l_{ij}) \hat{\mathbf{e}}_{ij}^e = \mathbf{0}, \quad i = 1, \dots, N.} \quad (8)$$

These are $3N$ scalar equations, of which $3N - 6$ are independent: V depends on the conformation only through the distances (2), so it is invariant under the three translations and three rotations, and along those six directions the gradient vanishes identically.

Two consequences of that invariance should be recorded now, because both matter later. First, $\mathbf{r}^e$ is determined only up to a rigid motion. Second, V is invariant under reflection as well, since reflecting a conformation leaves every d_{ij} unchanged; a chiral $\mathbf{r}^e$ therefore has a mirror image with the same energy, and the minimum of V is not unique. Below, $\mathbf{r}^e$ denotes one chosen minimiser, and every quantity derived from it is understood relative to that choice.

2.2 Strain

Let $\mathbf{r}^e$ be an equilibrium conformation and define, for every pair,

$$s_{ij} \equiv l_{ij} - d_{ij}(\mathbf{r}^e), \quad (9)$$

the amount by which the spring departs from its equilibrium length there. A spring with $s_{ij} > 0$ is compressed, one with $s_{ij} < 0$ is stretched, and one with $s_{ij} = 0$ is relaxed.

Substituting (9) into (5) evaluated at $\mathbf{r}^e$, and using $d_{ij}(\mathbf{r}^e) - l_{ij} = -s_{ij}$, gives the minimum energy

$$V(\mathbf{r}^e; \Lambda) = \frac{1}{2} \sum_{i < j} k(l_{ij}) s_{ij}^2. \quad (10)$$

The equilibrium condition (8) may be written in the same terms, $\sum_{j \neq i} k(l_{ij}) s_{ij} \hat{\mathbf{e}}_{ij}^e = \mathbf{0}$: the forces on each node cancel, which does not require the individual s_{ij} to vanish.

2.3 Frustration

A network is **frustrated** when $V(\mathbf{r}^e; \Lambda) > 0$, that is, when no conformation relaxes all its active springs at once. By (10) this holds whenever $s_{ij} \neq 0$ for at least one active pair.

The unfrustrated case is the special solution of (8) in which every term vanishes separately, $s_{ij} = 0$ for all active pairs. It is attainable: given any structure, setting $l_{ij} = d_{ij}$ for every pair makes that structure an equilibrium conformation with $V = 0$, and this is what a standard elastic network model does. But it is not the general solution. Equation (8) imposes $3N - 6$ independent conditions on $\frac{1}{2}N(N - 1)$ parameters, and for N of any size the system is underdetermined, so sets Λ with cancelling but individually nonzero s_{ij} are the generic case. As will be seen in Section 5, a mutation applied to an unfrustrated network produces a frustrated one.

3 The conformational distribution

3.1 The Boltzmann distribution

At temperature T the conformations are distributed according to

$$\rho(\mathbf{r}) = \frac{e^{-\beta V(\mathbf{r}; \Lambda)}}{Z}, \quad Z = \int e^{-\beta V(\mathbf{r}; \Lambda)} d\mathbf{r}, \quad \beta = \frac{1}{k_B T}, \quad (11)$$

where the integral runs over $\mathbb{R}^{3N}$. This is a configurational integral: the momenta have been left out. They contribute a factor that depends on the masses and on β but not on Λ , so it cancels from every difference between networks computed in Section 6, which is all we use.

3.2 The Hessian

The integral in (11) cannot be evaluated in closed form, because V is not quadratic in $\mathbf{r}$: the distances (2) are nonlinear functions of the coordinates. We therefore expand V about an equilibrium conformation. Writing $\mathbf{u} = \mathbf{r} - \mathbf{r}^e$,

$$V(\mathbf{r}; \Lambda) = V(\mathbf{r}^e; \Lambda) + \left. \frac{\partial V}{\partial \mathbf{r}} \right|_{\mathbf{r}^e} \cdot \mathbf{u} + \frac{1}{2} \mathbf{u}^T \mathbf{K} \mathbf{u} + O(\mathbf{u}^3), \quad \mathbf{K} = \left. \frac{\partial^2 V}{\partial \mathbf{r}^2} \right|_{\mathbf{r}^e}. \quad (12)$$

The linear term vanishes by (8).

To obtain $\mathbf{K}$ we need the second derivatives of the distances. Differentiating $\hat{\mathbf{e}}_{ij} = \mathbf{r}_{ij}/d_{ij}$ with respect to $\mathbf{r}_j$, using the product rule with $\partial \mathbf{r}_{ij}/\partial \mathbf{r}_j = -\mathbf{I}$ and $\partial d_{ij}^{-1}/\partial \mathbf{r}_j = -d_{ij}^{-2}(-\hat{\mathbf{e}}_{ij}^T) = d_{ij}^{-2} \hat{\mathbf{e}}_{ij}^T$,

$$\frac{\partial^2 d_{ij}}{\partial \mathbf{r}_i \partial \mathbf{r}_j} = \frac{\partial \hat{\mathbf{e}}_{ij}}{\partial \mathbf{r}_j} = -\frac{\mathbf{I}}{d_{ij}} + \frac{\mathbf{r}_{ij} \hat{\mathbf{e}}_{ij}^T}{d_{ij}^2} = \frac{\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T - \mathbf{I}}{d_{ij}}. \quad (13)$$

Now differentiate the gradient (7) with respect to $\mathbf{r}_j$ for $j \neq i$. Only the single term of the sum with that index depends on $\mathbf{r}_j$, and it is a product of $(d_{ij} - l_{ij})$ and $\hat{\mathbf{e}}_{ij}$, so

$$\mathbf{K}_{ij} = k(l_{ij}) \left[\frac{\partial (d_{ij} - l_{ij})}{\partial \mathbf{r}_j} \hat{\mathbf{e}}_{ij}^\top + (d_{ij} - l_{ij}) \frac{\partial \hat{\mathbf{e}}_{ij}}{\partial \mathbf{r}_j} \right]. \quad (14)$$

Substituting $\partial d_{ij} / \partial \mathbf{r}_j = -\hat{\mathbf{e}}_{ij}$ and (13), and then $d_{ij} - l_{ij} = -s_{ij}$ from (9),

$$\mathbf{K}_{ij} = k(l_{ij}) \left[-\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^\top - s_{ij} \frac{\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^\top - \mathbf{I}}{d_{ij}} \right], \quad (15)$$

that is, defining $g_{ij} \equiv s_{ij} / d_{ij}(\mathbf{r}^e)$,

$$\boxed{\mathbf{K}_{ij} = -k(l_{ij}) \left[\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^\top + g_{ij} (\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^\top - \mathbf{I}) \right], \quad i \neq j.} \quad (16)$$

The diagonal blocks follow from translational invariance. Displacing every node by the same vector $\mathbf{a}$ leaves V unchanged, so $\mathbf{K}$ annihilates that displacement, and summing the blocks in a row gives $\sum_j \mathbf{K}_{ij} \mathbf{a} = \mathbf{0}$ for every $\mathbf{a}$, whence

$$\mathbf{K}_{ii} = - \sum_{j \neq i} \mathbf{K}_{ij}. \quad (17)$$

Inactive pairs have $k(l_{ij}) = 0$ and contribute nothing, so $\mathbf{K}$ is $3N \times 3N$ however many pairs carry parameters.

3.3 Longitudinal and transverse stiffness

Equation (16) contains two projectors. Writing $\mathbf{P}_\perp = \mathbf{I} - \hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^\top$ for the projector onto the plane perpendicular to the pair, it may be rearranged as

$$\mathbf{K}_{ij} = -k(l_{ij}) \left[\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^\top - g_{ij} \mathbf{P}_\perp \right], \quad (18)$$

so a pair carries two stiffnesses rather than one: a longitudinal $k(l_{ij})$ resisting change of the distance, and a transverse $-k(l_{ij})g_{ij}$, doubly degenerate, resisting rotation of the pair at fixed distance.

The transverse stiffness is proportional to $g_{ij} = s_{ij} / d_{ij}$, and so vanishes for a relaxed spring and changes sign with the strain. A stretched spring ($s_{ij} < 0$) resists transverse displacement, as a taut string does. A compressed one ($s_{ij} > 0$) has negative transverse stiffness: displacing the two nodes sideways increases their separation, which a compressed spring favours. Note that it is the transverse stiffness that is negative, not $k(l_{ij})$, which is non-negative by (4).

Since $g_{ij} = 0$ for every pair of an unfrustrated network, the transverse term is absent there, and (16) reduces to the rank-one form of a standard elastic network model. Frustration enters the conformational distribution through g_{ij} and through nothing else.

3.4 The Gaussian approximation

Truncating (12) after the quadratic term gives

$$V(\mathbf{r}; \Lambda) \simeq V(\mathbf{r}^e; \Lambda) + \frac{1}{2} \mathbf{u}^\top \mathbf{K} \mathbf{u}, \quad (19)$$

and hence, from (11), a Gaussian distribution of conformations centred on $\mathbf{r}^e$,

$$\rho(\mathbf{r}) \simeq \frac{1}{Z} e^{-\beta V(\mathbf{r}^e; \Lambda)} \exp\left(-\frac{\beta}{2} \mathbf{u}^\top \mathbf{K} \mathbf{u}\right). \quad (20)$$

$\mathbf{K}$ is symmetric, and positive semi-definite because $\mathbf{r}^e$ is a minimum. Let its eigenvalues be λ_n with eigenvectors $\mathbf{q}_n$. At least six eigenvalues vanish, corresponding to the three translations and three rotations along which V does not vary.

Whether there are exactly six is a property of the network, not a given. A displacement $\mathbf{u}$ is a null direction of $\mathbf{K}$ when it changes no active distance to first order, that is when $\hat{\mathbf{e}}_{ij} \cdot (\mathbf{u}_i - \mathbf{u}_j) = 0$ for every active pair. A network for which the only such displacements are the six rigid motions is *infinitesimally rigid*, and then $\text{rank } \mathbf{K} = 3N - 6$. Connectivity does not suffice: a node joined to the rest by a single active spring can swing about it, and one joined by two can rotate about the axis through them, in both cases without changing any active distance. Each such freedom is a further null direction. In what follows the networks are assumed infinitesimally rigid, and we write $\sum'$ and $\prod'$ for sums and products over the $3N - 6$ nonzero eigenvalues.

The distribution (20) is flat along the six null directions, which is the rigid-body freedom of the whole network, and we factor it out: displacements are taken in the complement, where the covariance is

$$\boldsymbol{\Sigma} = \beta^{-1} \mathbf{K}^+, \quad \mathbf{K}^+ = \sum_n' \lambda_n^{-1} \mathbf{q}_n \mathbf{q}_n^T, \quad (21)$$

with $\mathbf{K}^+$ the pseudo-inverse obtained by inverting the nonzero eigenvalues and annihilating the null directions.

4 Thermodynamic properties

4.1 The partition function

In the quadratic approximation (19) the integral (11) factorises in the eigenbasis of $\mathbf{K}$. Writing $\mathbf{u} = \sum_n y_n \mathbf{q}_n$, the quadratic form is $\sum_n \lambda_n y_n^2$, and the integral over each of the $3N - 6$ non-null directions is a standard Gaussian,

$$\int_{-\infty}^{\infty} e^{-\beta \lambda_n y_n^2 / 2} dy_n = \left(\frac{2\pi}{\beta \lambda_n} \right)^{1/2}. \quad (22)$$

Hence

$$Z = e^{-\beta V(\mathbf{r}^e; \Lambda)} \prod_n' \left(\frac{2\pi}{\beta \lambda_n} \right)^{1/2}, \quad \ln Z = -\beta V(\mathbf{r}^e; \Lambda) + \frac{1}{2} \sum_n' \ln \frac{2\pi}{\beta \lambda_n}. \quad (23)$$

4.2 Free energy, internal energy, entropy

The Helmholtz free energy is

$$A = -\beta^{-1} \ln Z = V(\mathbf{r}^e; \Lambda) - \frac{1}{2\beta} \sum_n' \ln \frac{2\pi}{\beta \lambda_n}. \quad (24)$$

The internal energy follows by differentiating (23), using $\partial \ln \lambda_n / \partial \beta = 0$,

$$U = -\frac{\partial \ln Z}{\partial \beta} = V(\mathbf{r}^e; \Lambda) + \frac{3N - 6}{2\beta}, \quad (25)$$

which is the minimum energy plus $\frac{1}{2} k_B T$ per non-rigid degree of freedom, as equipartition requires. The entropy is $S = (U - A)/T$, so

$$TS = U - A = \frac{1}{2\beta} \sum_n' \left[\ln \frac{2\pi}{\beta \lambda_n} + 1 \right]. \quad (26)$$

Soft directions, those of small λ_n , contribute most: the network gains entropy by being easy to deform.

Two comments on (24) and (26). The logarithm takes a dimensional argument, so the numerical values of A and TS depend on the units chosen for length, and A may fall below $V(\mathbf{r}^e; \Lambda)$. Neither is meaningful alone. What is meaningful is a difference between two networks at the same temperature, where the units cancel; those are the quantities of Section 6. The same applies to the momentum factor dropped in (11), and to the rigid-body volume factored out of (21): both are common to the two networks and cancel.

Open question. Equations (24) and (26) rest on the quadratic truncation (19). How good it is for a frustrated network, where the transverse term makes $\mathbf{K}$ differ from the standard form, has not been examined here.

5 Reference and mutant

5.1 The reference network

Everything above applies to any network. We now fix one, the *reference*, with parameters $\Lambda^{\text{ref}} = \{l_{ij}^{\text{ref}}\}$ and potential

$$V_{\text{ref}}(\mathbf{r}) \equiv V(\mathbf{r}; \Lambda^{\text{ref}}) = \frac{1}{2} \sum_{i < j} k(l_{ij}^{\text{ref}}) \left(d_{ij}(\mathbf{r}) - l_{ij}^{\text{ref}} \right)^2. \quad (27)$$

Write $k_{ij}^{\text{ref}} = k(l_{ij}^{\text{ref}})$ for its force constants, $\mathbf{r}_{\text{ref}}^e$ for an equilibrium conformation obtained from (8), and

$$s_{ij} = l_{ij}^{\text{ref}} - d_{ij}(\mathbf{r}_{\text{ref}}^e) \quad (28)$$

for its strains. Its Hessian $\mathbf{K}^{\text{ref}}$, spectrum $\{\lambda_n^{\text{ref}}\}$, and thermodynamic properties A_{ref} , U_{ref} , S_{ref} follow from Sections 3 and 4. The reference may be frustrated or not; nothing below assumes either.

5.2 Mutation

A mutation at site m perturbs the equilibrium lengths of the springs that connect m to the rest of the network, and leaves all others unchanged:

$$l_{ij}^{\text{mut}} = l_{ij}^{\text{ref}} + \delta l_{ij}, \quad \delta l_{ij} = 0 \quad \text{unless } i = m \text{ or } j = m. \quad (29)$$

The perturbations δl_{ij} are drawn from a distribution with zero mean and variance σ^2 , independently for each affected pair; σ sets the size of the mutation.

The mutant potential is (5) evaluated with these parameters,

$$V_{\text{mut}}(\mathbf{r}) \equiv V(\mathbf{r}; \Lambda^{\text{mut}}) = \frac{1}{2} \sum_{i < j} k(l_{ij}^{\text{ref}} + \delta l_{ij}) \left(d_{ij}(\mathbf{r}) - l_{ij}^{\text{ref}} - \delta l_{ij} \right)^2. \quad (30)$$

Comparing (27) and (30): the mutation changes both the equilibrium lengths and, through (4), the force constants. Define

$$\delta k_{ij} \equiv k(l_{ij}^{\text{ref}} + \delta l_{ij}) - k(l_{ij}^{\text{ref}}), \quad \text{so} \quad k_{ij}^{\text{mut}} = k_{ij}^{\text{ref}} + \delta k_{ij}. \quad (31)$$

This is an exact difference of two values of k , not a differential: no assumption is made that δl_{ij} is small or that k is differentiable. A pair whose equilibrium length crosses from a region where

$k > 0$ to one where $k = 0$, or the reverse, has δk_{ij} of the size of k itself. Contacts are created and destroyed here, in the parameters, and not by any change in conformation.

The mutant has its own equilibrium conformation $\mathbf{r}_{\text{mut}}^e$, obtained from (8) with the mutant parameters, its own strains $s_{ij}^{\text{mut}} = l_{ij}^{\text{mut}} - d_{ij}(\mathbf{r}_{\text{mut}}^e)$, its own Hessian $\mathbf{K}^{\text{mut}}$ and spectrum $\{\lambda_n^{\text{mut}}\}$, and its own $A_{\text{mut}}, U_{\text{mut}}, S_{\text{mut}}$. Note that $\mathbf{r}_{\text{mut}}^e$ is obtained by minimising (30) itself, not by displacing $\mathbf{r}_{\text{ref}}^e$ along a response to the perturbation.

Even if the reference is unfrustrated, the mutant will in general be frustrated: the perturbed lengths l_{ij}^{mut} need not be simultaneously realisable by any conformation, so the mutant settles at a compromise with $s_{ij}^{\text{mut}} \neq 0$.

6 Mutational changes

6.1 Three energy differences

Two networks and two conformations give three energy differences worth naming. The first is the cost of changing the parameters at fixed conformation,

$$\Delta V_{\text{stress}} = V_{\text{mut}}(\mathbf{r}_{\text{ref}}^e) - V_{\text{ref}}(\mathbf{r}_{\text{ref}}^e), \quad (32)$$

the second the energy released as the mutant relaxes to its own equilibrium,

$$\Delta V_{\text{relax}} = V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) - V_{\text{mut}}(\mathbf{r}_{\text{ref}}^e), \quad (33)$$

and their sum is the difference between the two minima,

$$\Delta V_{\text{min}} = V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) - V_{\text{ref}}(\mathbf{r}_{\text{ref}}^e) = \Delta V_{\text{stress}} + \Delta V_{\text{relax}}. \quad (34)$$

Since $\mathbf{r}_{\text{mut}}^e$ minimises V_{mut} , we have $V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) \leq V_{\text{mut}}(\mathbf{r})$ for every $\mathbf{r}$, and in particular for $\mathbf{r} = \mathbf{r}_{\text{ref}}^e$, so

$$\Delta V_{\text{relax}} \leq 0. \quad (35)$$

6.2 Decomposition of the stress energy

ΔV_{stress} can be written out in closed form. We do so in steps.

Step 1: separate the pairs. Both (27) and (30) are sums over the same index set, all $\frac{1}{2}N(N-1)$ pairs, so their difference is a sum of pair terms,

$$\Delta V_{\text{stress}} = \sum_{i < j} \left[\frac{1}{2} k_{ij}^{\text{mut}} (d_{ij}^e - l_{ij}^{\text{mut}})^2 - \frac{1}{2} k_{ij}^{\text{ref}} (d_{ij}^e - l_{ij}^{\text{ref}})^2 \right], \quad d_{ij}^e \equiv d_{ij}(\mathbf{r}_{\text{ref}}^e). \quad (36)$$

This is where carrying a parameter for every pair, active or not, does its work: no pair appears in one sum and not the other, and a contact created or destroyed by the mutation is carried by δk_{ij} rather than by a change in the range of summation.

Step 2: eliminate the distance. By the definition of the reference strain (28), $d_{ij}^e = l_{ij}^{\text{ref}} - s_{ij}$, so the two brackets become

$$d_{ij}^e - l_{ij}^{\text{ref}} = -s_{ij}, \quad d_{ij}^e - l_{ij}^{\text{mut}} = -s_{ij} - \delta l_{ij} = -(s_{ij} + \delta l_{ij}). \quad (37)$$

The conformation has now disappeared from the problem: what remains is expressed entirely in the reference strains and the perturbation.

Step 3: substitute the force constants. Using (31), the pair term is

$$\frac{1}{2}(k_{ij}^{\text{ref}} + \delta k_{ij})(s_{ij} + \delta l_{ij})^2 - \frac{1}{2}k_{ij}^{\text{ref}} s_{ij}^2. \quad (38)$$

Step 4: split off the change in stiffness. Grouping the two terms in k_{ij}^{ref} and keeping the δk_{ij} term apart,

$$= \frac{1}{2}k_{ij}^{\text{ref}}[(s_{ij} + \delta l_{ij})^2 - s_{ij}^2] + \frac{1}{2}\delta k_{ij}(s_{ij} + \delta l_{ij})^2. \quad (39)$$

The second piece is already in final form. No expansion has been made in obtaining it, and none is possible: δk_{ij} is not assumed small.

Step 5: expand the square. The bracket is a difference of two squares,

$$(s_{ij} + \delta l_{ij})^2 - s_{ij}^2 = 2s_{ij}\delta l_{ij} + \delta l_{ij}^2, \quad (40)$$

which is a finite algebraic identity, not a truncated expansion: it is exact for any δl_{ij} .

Step 6: collect. Summing over pairs,

$$\Delta V_{\text{stress}} = \underbrace{\frac{1}{2} \sum_{i < j} k_{ij}^{\text{ref}} \delta l_{ij}^2}_{(a)} + \underbrace{\sum_{i < j} k_{ij}^{\text{ref}} s_{ij} \delta l_{ij}}_{(b)} + \underbrace{\frac{1}{2} \sum_{i < j} \delta k_{ij} (s_{ij} + \delta l_{ij})^2}_{(c)}. \quad (41)$$

This holds for any k , any mutation size, and any reference network, frustrated or not.

6.3 The three terms

Term (a) comes from the δl_{ij}^2 of the expanded square. It is non-negative, quadratic in the size of the mutation, and independent of the state of the reference.

Term (b) is the cross term $2s_{ij}\delta l_{ij}$ of the same square. It is proportional to the reference strain and is the only term of the three that can be negative: a mutation that shortens a stretched spring, or lengthens a compressed one, releases energy. It vanishes when $s_{ij}\delta l_{ij} = 0$ for every pair, which is to say when the reference is unstrained on the pairs the mutation touches. In particular it vanishes for any mutation of an unfrustrated reference, and it does not vanish in general.

Term (c) is the whole of the contribution from the change in force constants. It vanishes when $\delta k_{ij}(s_{ij} + \delta l_{ij})^2 = 0$ pairwise, in particular when k is constant over the range the mutation explores. It is the term that a model with a fixed contact map does not have.

Terms (b) and (c) are what distinguish (41) from the energy cost of mutating an unfrustrated network with fixed force constants, which is (a) alone. Neither is a correction of higher order in the mutation size: both are exact contributions of the same standing as (a), which happen to vanish in that case.

6.4 Differences and energies

ΔV_{stress} is a difference of two potentials, and is not the same as the absolute energy $V_{\text{mut}}(\mathbf{r}_{\text{ref}}^e)$ of the mutant network at the reference conformation. By (32) the two differ by $V_{\text{ref}}(\mathbf{r}_{\text{ref}}^e)$, the strain energy the reference already carries, which by (10) is zero for an unfrustrated reference and positive otherwise.

6.5 Changes in the thermodynamic properties

Applying Section 4 to both networks and subtracting, the internal energy difference follows from (25),

$$\Delta U = U_{\text{mut}} - U_{\text{ref}} = V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) - V_{\text{ref}}(\mathbf{r}_{\text{ref}}^e) = \Delta V_{\text{min}}, \quad (42)$$

the equipartition terms cancelling. This step, and the entropy difference below, require the two networks to have the same number of non-null directions. A point mutation changes the parameters of one site and neither adds nor removes a node, so both Hessians are $3N \times 3N$; and if both networks are infinitesimally rigid in the sense of Section 3, both have $3N - 6$ nonzero eigenvalues. For the entropy, write (26) as

$$TS = \frac{1}{2\beta} \left[(3N - 6) \left(\ln \frac{2\pi}{\beta} + 1 \right) - \sum_n' \ln \lambda_n \right], \quad (43)$$

in which only the last term depends on the network. Both networks have the same number of non-null directions, so on subtracting, the constant term cancels and

$$T\Delta S = \frac{1}{2\beta} \left[\sum_n' \ln \lambda_n^{\text{ref}} - \sum_n' \ln \lambda_n^{\text{mut}} \right] = \frac{1}{2\beta} \ln \frac{\prod_n' \lambda_n^{\text{ref}}}{\prod_n' \lambda_n^{\text{mut}}}. \quad (44)$$

The two spectra enter only through these products. No correspondence between the modes of the reference and those of the mutant is used, and none is available: the eigenvectors of $\mathbf{K}^{\text{ref}}$ and $\mathbf{K}^{\text{mut}}$ are different sets of directions, and the eigenvalues are not paired by any property of the model.

The free energy difference is then

$$\Delta A = \Delta U - T\Delta S = \Delta V_{\text{min}} - \frac{1}{2\beta} \ln \frac{\prod_n' \lambda_n^{\text{ref}}}{\prod_n' \lambda_n^{\text{mut}}}. \quad (45)$$

The dimensional ambiguity noted after (26) has cancelled: (44) and (45) depend only on the ratio of the two products. In the protein literature the quantity (45) is what is meant by $\Delta\Delta G$ for the folded state.

The products $\prod_n' \lambda_n$ are the pseudo-determinants of the two Hessians, so (44) may be computed without diagonalising either matrix.

Equations (34) and (44) separate the two ways a mutation changes the free energy. It shifts the minimum of the potential, which (41) decomposes; and it changes the spectrum of $\mathbf{K}$, which by (16) depends on the strains through g_{ij} and on the contact map through $k(l_{ij})$. A mutation that leaves the minimum energy untouched can still change the free energy, through the spectrum alone.

7 The step function

The results above hold for any k . The common choice is a step at a cutoff distance,

$$k(l) = k_0 \mathbb{1}[l \leq d_{\text{max}}], \quad (46)$$

under which a pair is either a spring of strength k_0 or nothing at all. Two consequences follow from the results above.

First, V is discontinuous in the parameters. If a mutation carries l_{ij}^{ref} across d_{max} then $\delta k_{ij} = \pm k_0$, and by term (c) of (41) the energy changes by $\pm \frac{1}{2} k_0 (s_{ij} + \delta l_{ij})^2$ however small δl_{ij} is.

Second, and for the same reason, term (c) cannot be treated as small when δl_{ij} is small. This is the sense in which the contact map resists a perturbative treatment: δk_{ij} is not controlled by δl_{ij} .

A continuous k , for instance a sigmoid of width w centred on $d_{\max}$, makes V continuous in Λ and δk_{ij} small whenever δl_{ij} is small. The cost is that no pair is exactly inactive, so the contact map becomes a threshold imposed on a continuum rather than a property of the network.

8 Open questions and work to come

The following are not settled here.

- The quadratic truncation (19) is the only approximation in the document. Its accuracy for a frustrated network has not been examined.
- The magnitudes of (a), (b) and (c) in (41) for a real protein are not known. The decomposition says when each vanishes, not how large each is.
- How large s_{ij} may become before a compressed spring, with its negative transverse stiffness (18), ceases to represent a residue contact.
- A mutation can cost the network its infinitesimal rigidity. The perturbation (29) acts on every spring of the mutated site, and if enough of their equilibrium lengths leave the range where $k > 0$, the site is left attached by one or two springs and acquires a rotational freedom that changes no active distance. The mutant Hessian then has more than six null directions, the counts in Section 6 no longer match, and neither $\Delta U = \Delta V_{\min}$ nor (44) holds as written: an uncanceled term in $\ln(2\pi/\beta)$ survives, so ΔA is no longer independent of the units of length. How often this happens at a given mutation size and cutoff is an empirical question, and what the free energy difference should mean when it does is not settled here.
- Whether a frustrated network is a better model of a native structure than an unfrustrated one, and if so how the l_{ij} should be chosen, given that (8) does not determine them.

To be added: the distribution of structures generated by repeated mutation, and the entropic contribution to $\Delta\Delta G$ along a substitution process.